Demand Forecasting for Public Library Services

This paper examines how a four-site library network can forecast demand for lending, reference and programming services using circulation logs, visitor counts and program registrations. We treat forecasts as decision aids rather than verdicts, and we separate simulated observations from causal claims throughout. The analysis covers data preparation, model comparison, scenario interpretation and the governance duties that keep forecasts accountable to branch staff and residents.

A Study Scope and Forecasting Questions

The study addresses a practical question: how many loans, visits and program seats should each branch plan for in the coming cycle? We frame forecasting as a planning instrument for staffing, collection rotation and room booking, not as a performance rating. The scope covers the main circulating collections and public programs across the participating sites, while excluding interlibrary transfers and one-off outreach events that follow their own scheduling logic.

A.1 Data Sources and Preparation

Records come from circulation systems, door counters and registration logs covering 2022 through 2025. We harmonize branch identifiers, remove duplicate checkouts, and flag closure days and system outages before aggregation. Weekly series are constructed so that holiday spikes are visible rather than smoothed away, and every exclusion is logged so that reviewers can reconstruct the prepared dataset from the raw extracts.

Supporting records are presented in Figure 1.

Rule	Action	Tag	Reviewer
Test scans	Remove	R1	Second pass
Bulk loans	Exclude	R2	Supervisor
Duplicates	Collapse	R3	Sample
Outages	Flag	R4	Data lead

Table 1: Cleaning rule checks applied to circulation extracts

A.1.1 Cleaning Rules for Circulation Records



Figure 1: Data preparation steps and their review owners

Duplicate scans, staff test checkouts and bulk class loans are removed under written rules agreed with branch supervisors. Each removal is tagged with a reason code, and a sample of flagged rows is re-checked by a second reviewer. The cleaning log records the rule version applied, so a later analyst can reproduce the same dataset and see exactly which records changed and why.

Supporting records are presented in Table 1.

A.1.2 Handling Closures and Seasonal Gaps

Closure days, renovation periods and holiday weeks create gaps that distort naive averages. We mark these intervals explicitly and model them as known absences rather than missing demand. Where a gap exceeds a

full week, the branch series is re-based on the surrounding comparable weeks, and the adjustment is documented alongside the original counts so that planners can judge how much of a forecast rests on imputed values.

A.2 Constructing Weekly Demand Series

Weekly aggregates combine loans, visits and program registrations into separate series rather than a single composite index. Keeping the series apart preserves interpretability: a rise in visits without a matching rise in loans suggests different service use. Each series carries a data-quality flag, and weeks below a completeness threshold are excluded from model fitting but retained in the audit trail for transparency.

B Model Comparison and Validation

We compare a seasonal baseline, a regression with calendar and weather covariates, and a gradient-boosted model across the prepared series. Validation uses rolling-origin splits so that each model is tested only on weeks after its training window. Accuracy is reported per branch and per service line, because aggregate scores can hide a branch where a model consistently under-predicts weekend demand.

B.1 Evaluation Metrics and Error Patterns

Forecast error is summarized with mean absolute percentage error and a bias measure that reveals systematic over- or under-prediction. Errors cluster around school holiday transitions and program launch weeks, which suggests calendar features deserve more weight than weather variables. We also report error by branch size, since small branches show wider relative error even when absolute differences are modest.

Supporting records are presented in Table 4.

B.1.1 Backtesting Across Branches

Backtests replay 12 weeks of history for each of the 4 sites, refitting models at fixed intervals. Results show that the seasonal baseline is competitive for loans but weaker for program registrations, which respond to announcement timing. These are simulated backtest observations, not evidence that any single model causes better service outcomes; they indicate where model choice matters most for planning.



Figure 2: Interpretation guide for forecast intervals by service

B.1.2 Interpreting Forecast Intervals

Point forecasts are presented with prediction intervals so planners can see the plausible range. A branch planning staffing against the upper bound will over-staff on average, while planning against the lower bound risks queues at peak hours. We recommend choosing the interval width according to the cost of being wrong for each service, and we label intervals as simulated estimates rather than guarantees.

Supporting records are presented in Figure 2.

Duty	Role	Timing	Output
Assumption check	Supervisor	Pre-roster	Memo
Error review	Analyst	Quarterly	Report
Model change	Data lead	As needed	Rationale
Decision record	Manager	Each cycle	Log entry

Table 2: Review duties and cadence for forecast governance

B.2 Governance and Implementation Duties

Forecasts enter branch planning only after a review meeting where supervisors check assumptions, note local events and record dissent. The review produces a decision memo with owners and review dates, and model changes require a documented rationale. This governance keeps forecasting accountable: staff know which numbers drive rosters, and residents can ask how service levels were set without confronting an unexplained algorithm.

Supporting records are presented in Table 2.

Supporting records are presented in Table 3.

Check	Owner	Status	Evidence
Calendar match	Supervisor	Open	Closure list
Interval brief	Analyst	Open	Plain summary
Owner named	Manager	Open	Memo
Shadow run	Data lead	Open	Comparison

Table 3: Implementation readiness checks before forecasts drive rosters

Branch supervisors should confirm that holiday calendars used in fitting match the actual closure schedule; a mismatch quietly shifts seasonal effects and inflates error in the weeks surrounding each break.

When a counter fails for part of a day, the affected week should be flagged rather than scaled up, because scaling assumes a stable visitor pattern that the outage itself may have disturbed.

Model documentation should name the training window, feature set and validation split so that a later analyst can reproduce results without reconstructing decisions from memory or informal notes.

Program registration series often lead actual attendance; planners should keep both measures visible so that forecasted demand reflects seats offered as well as seats filled.

Rolling-origin validation is preferred to a single holdout because library demand shifts gradually and one split can flatter a model that performs poorly in other periods.

Where two models disagree sharply for a branch, the review meeting should record both ranges and the reasoning behind the chosen planning figure rather than averaging silently.

Small branches need wider intervals; treating their forecasts with the same precision as large sites invites overconfidence in staffing and collection decisions.

Weather covariates add little explanatory value in our simulated tests, so teams should weigh whether the data collection burden is justified before retaining them.

Renovation periods should be excluded from fitting but retained in reporting, since planners still need to know what demand looked like before and after the disruption.

Forecast review should occur before roster deadlines, not after, so that adjustments can still influence staffing rather than merely annotate past choices.

Any change to cleaning rules should trigger a refit and a short note explaining the expected effect, because rule changes can move error metrics without any real change in demand.

Reference desk queries are poorly captured by

door counters alone; a separate tally or sampling scheme improves visibility into demand that never reaches the circulation system.

Criterion	Baseline	Regression	Boosted
Seasonal fit	Strong	Moderate	Strong
Calendar use	None	Yes	Yes
Interpretability	High	High	Low
Setup effort	Low	Medium	High

Table 4: Model comparison criteria used in validation